

DL for ENG

## LECTURE 11.2

# Dynamics, Energy

The physics comes back as ordinary differential equations, not partial ones. Three models say how the car moves, what it cannot do, and what the manoeuvre costs.

# and Efficient Driving

Boris Todorov

With support from Research Assistant Noman Khan - [nomank@energy.aau.dk](mailto:nomank@energy.aau.dk)

Aalborg University · [ret@et.aau.dk](mailto:ret@et.aau.dk) · Deep Learning for Engineering · 2026

Kinematic bicycle model · DC motor and 18650 pack · INA219 telemetry





# What You Will Be Able To Do

## THE GOALS, IN PLAIN WORDS

- **write** — the kinematic bicycle model, and name its single parameter
- **calibrate** — a steering map by measurement, not by assumption
- **state** — the grip limit, and why braking and cornering share it
- **choose** — between a soft penalty and a hard layer, and defend the choice
- **explain** — why the energy-optimal speed is not the slowest speed
- **connect** — the pack model here to the one you built in L10.1

## WHAT TODAY ASSUMES

- |            |                                   |
|------------|-----------------------------------|
| from L11.1 | the car, and a working policy     |
| from L7.1  | hard against soft enforcement     |
| from L10.1 | the equivalent-circuit pack model |

## PHYSICS WITHOUT A PDE

Ordinary differential equations, algebraic constraints, and an objective. The machinery of L7 applies unchanged.

## AND ONE UNGLAMOROUS SLIDE

The steering calibration decides whether every later result means anything.

# Where the Physics Comes Back

THE CLAIM, STATED PRECISELY

L11.1 had no governing equation. L11.2 has three — kinematics, a friction limit, and an energy model — and none of them is a PDE. They are ordinary differential equations in time, and they enter the loss exactly as a PDE residual did in L7. The machinery is unchanged; only the operator is.

|                                | L11.1                         | L11.2                               |
|--------------------------------|-------------------------------|-------------------------------------|
| What constrains the model      | labelled images               | an ODE residual and a grip limit    |
| What you can verify against    | held-out data, then the floor | a measured power trace              |
| What limits you                | latency and data              | model error and unmodelled friction |
| Typical failure                | the car hits a wall           | the car is slow, or flat, or both   |
| The instrument that decides it | a stopwatch                   | the INA219                          |

# Yes, This Is Still a PINN — Mostly

## WHERE THE PART 2 MACHINERY APPLIES

- identifying the vehicle parameters is a PINN inverse problem, with ODEs instead of PDEs
- a physics residual in the loss, autograd derivatives, coefficients made trainable
- **optimising the lap is a step beyond** — a non-zero objective, with the physics as a constraint
- so a PINN drives residuals to zero; here you minimise energy subject to them being zero

## WHICH PART 1 IDEAS ARE IN USE

|            |                                |
|------------|--------------------------------|
| from L7.1  | the composite residual loss    |
| from L6.1  | autograd, Adam, L-BFGS         |
| from L10.2 | physics inside an optimisation |
| from L11.1 | the policy being optimised     |

## WHAT IS AND IS NOT A PINN HERE

Identification is one, and the lap optimisation is optimal control through a differentiable model.

## AND THAT IS THE POINT

A differentiable model is one you can optimise through.

# The Kinematic Bicycle Model

## FOUR STATES, TWO INPUTS

$$\dot{x} = v \cos \theta, \quad \dot{y} = v \sin \theta, \quad \dot{\theta} = \frac{v}{L} \tan \delta, \quad \dot{v} = a$$

$(x, y)$  the position,  $\theta$  the heading  
 $v$  the speed,  $\delta$  the steering angle  
 $L$  the wheelbase,  $a$  the acceleration

- collapse the two front wheels onto one, the two rear onto one
- state is position, heading and speed; inputs are steering angle and acceleration
- the only vehicle parameter is the wheelbase
- mass, tyre stiffness and weight transfer are all absent

### THE ONE LINE TO REMEMBER

A steering command buys curvature, not yaw rate. Everything on slides 6, 15 and 17 follows from that sentence.

## CURVATURE AND TURN RADIUS

$$\kappa = \frac{\tan \delta}{L} \quad R = \frac{L}{\tan \delta} \quad R_{\min} = \frac{L}{\tan \delta_{\max}}$$

With  $L \approx 0.17$  m and full lock near  $30^\circ$ , the tightest circle the car can describe is about 0.29 m in radius — roughly one and a half car lengths.

## THE SLIP-ANGLE REFINEMENT

$$\beta = \arctan \left( \frac{l_r \tan \delta}{L} \right)$$

Referencing the centre of mass introduces a slip angle  $\beta$ . At JetRacer speeds the correction is smaller than your measurement error in  $L$ .

# What the Model Assumes, and When It Breaks

## No lateral slip

*assumes*

Each wheel rolls in the direction it points. The tyre generates whatever side force is required, instantly and for free.

Holds while lateral acceleration stays well below the friction limit — which is exactly the regime slide 6 defines.

## No weight transfer

*assumes*

Load stays evenly distributed under braking and cornering, so grip is constant at all four contact patches.

With real springs and a high pack this car rolls and pitches visibly. It shows up as understeer on turn-in, and the model will not predict it.

## Instant steering

*assumes*

The servo reaches the commanded angle immediately; there is no actuator lag and no rate limit.

The servo needs of order 100 ms for a large step. At a 20 Hz loop that is two commands, and it matters when you fit parameters.

## Planar and rigid

*assumes*

No suspension travel, no ground undulation, no chassis flex, and a perfectly level floor.

Knowingly false: the Pro has adjustable oil-filled shocks and independent suspension on all four corners. Ignored, not absent.

## Speed is commanded

*assumes*

Longitudinal acceleration  $a$  is treated as a free input the drivetrain can supply on demand.

This is the assumption L11.2 removes. Slides 8 to 10 replace  $a$  with a motor, a pack, and a torque the car actually has.

*The kinematic model is the correct model for this vehicle at this speed, and the fifth row is the only assumption L11.2 discards.*

# From Command to Steering Angle

- the model takes an angle in radians; the car takes a normalised servo command
- the map between them is not linear, because linkages have geometry
- **so measure it** — drive steady circles at fixed commands and invert
- there is also an offset: a car that drifts on zero command has a trim error
- fit a polynomial or a lookup table, not a neural network

*Twenty minutes of calibration at the start of the session. Every team that skips it spends longer than that later.*

## THE CALIBRATION DRIVE

- ① **Fix throttle low** constant, slow, well below the grip limit
- ② **Fix the steering command** one value, held for a full circle
- ③ **Measure the circle** chalk the floor, or time a full revolution
- ④ **Invert the geometry**  $\delta = \arctan(L / R)$  for that command
- ⑤ **Repeat, both directions** five commands each way; the car is not symmetric

## WHY THIS SLIDE EXISTS

Every later result depends on it. A model fitted through an uncalibrated steering map identifies a wrong wheelbase, a wrong friction coefficient, and an energy curve wrong by a constant factor.



# What the Car Physically Cannot Do

## THE CORNERING LIMIT

$$a_y = \frac{v^2}{R} \leq \mu g \quad \Rightarrow \quad v_{\max}(R) = \sqrt{\mu g R}$$

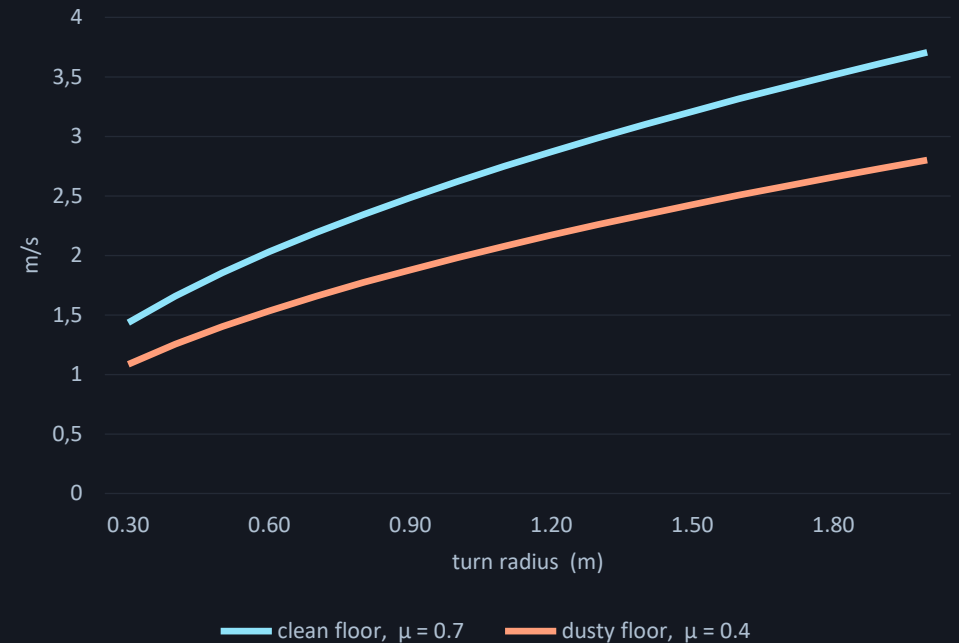
$a_y$  the lateral acceleration,  $R$  the turn radius  
 $\mu$  the friction coefficient,  $g$  gravity

- lateral acceleration in a steady turn is  $v^2$  over  $R$
- the contact patches can supply no more than  $\mu g$
- $\mu$  is a property of your floor and your tyres together, and it changes
- combined loading is stricter** — braking and cornering share one friction budget

$$\sqrt{a_x^2 + a_y^2} \leq \mu g$$

*measure  $\mu$  once per session — it is not a constant of nature*

## MAXIMUM SPEED AGAINST TURN RADIUS



A 0.5 m corner caps the car at 1.9 m/s on a clean floor and 1.4 m/s on a dusty one. L11.1 assumed 2 m/s throughout.

# Two Ways to Enforce a Constraint

## YOU HAVE SEEN THIS BEFORE

This is the soft-versus-hard distinction from L7.1, transplanted. There it was a boundary condition; here it is a grip limit. The argument for each is identical, and so is the failure mode of choosing wrongly.

### SOFT — A PENALTY IN THE LOSS

$$\mathcal{L}_{\text{con}} = \frac{1}{N} \sum_k [\max(0, a_{y,k} - \mu g)]^2$$

$a_{y,k}$  the lateral acceleration at sample  $k$ ,  $\mu g$  the limit

- a soft penalty is zero inside the feasible set and quadratic outside
- **it is a preference, not a guarantee** — the optimiser will trade a small violation
- a hard constraint squashes the output through a bounded function

### HARD — A LAYER THE POLICY CANNOT PASS

$$\delta = \delta_{\max} \tanh(\hat{\delta}), \quad v = v_{\max}(R) \sigma(\hat{v})$$

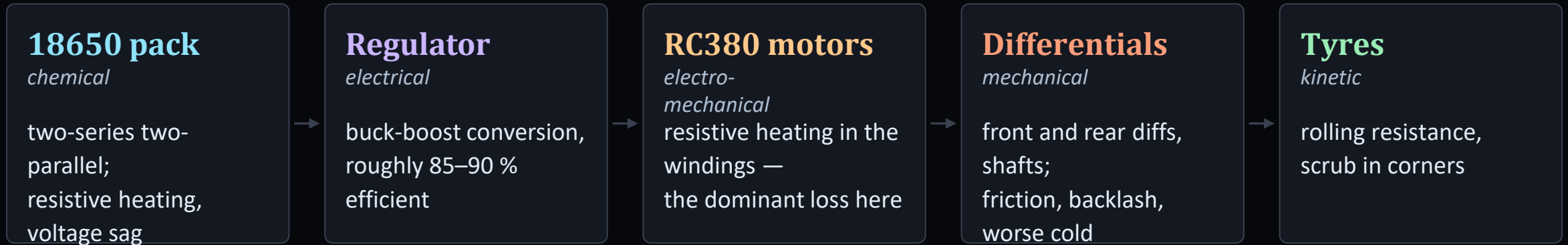
$\hat{\delta}, \hat{v}$  the raw outputs  
 $\sigma$  the logistic function

Squash the raw network output through a bounded function scaled by the limit. The steering command cannot exceed full lock and the speed command.

*Enforce the grip limit hard and the smoothness preference soft.*

# Where the Joules Actually Go

FROM CELL TO CONTACT PATCH — EVERY STAGE TAKES A CUT



**The Nano is a stage too — and it never stops drawing.**

- **the compute load** — a few watts, moving or not
- board, camera and display, drawing continuously
- **aerodynamic drag** — negligible at 2 m/s
- in a full-size electric vehicle both facts reverse
- so road-vehicle intuition gives the wrong optimal speed

## THE INVERTED INTUITION

In a full-size electric vehicle aerodynamic drag dominates at speed and the auxiliary load is a rounding error. On a JetRacer both reverse: drag is negligible, the constant compute load is not, and the largest single loss is resistive heating in the windings.

# The Brushed DC Motor, in Three Equations

## TORQUE, BACK-EMF, AND THE VOLTAGE BALANCE

$$\tau = k_t i, \quad e = k_e \omega, \quad V = iR_a + k_e \omega$$

$\tau$  torque,  $i$  current,  $e$  back EMF,  $\omega$  speed  
 $k_t, k_e$  the motor constants,  $R_a$  armature resistance

- torque is proportional to current; back-EMF is proportional to speed
- the applied voltage splits between resistance and back-EMF

## THE POWER SPLIT

$$P_{\text{elec}} = Vi = \underbrace{i^2 R_a}_{\text{heat}} + \underbrace{\tau \omega}_{\text{mechanical}}$$

- some of the input power turns the wheels; the rest becomes heat
- copper loss is quadratic in current, so quadratic in torque

## LONGITUDINAL DYNAMICS

$$m\dot{v} = \frac{G k_t i}{r_w} - C_{rr}mg - \frac{1}{2}\rho C_d A v^2$$

Gear ratio and wheel radius turn motor torque into tractive force, and enter only as the lumped constant above. The aerodynamic term you may keep only to show it does not matter.

**RC380 BRUSHED, AND NO ENCODER ON IT**  
 no encoder fitted

**traction constant**

**armature resistance**

**rolling coefficient**

the Pro has none — speed must come from lap timing or from vision  
 torque constant, final drive and wheel radius, only ever as one winding, brush and wiring product together; it rises as the motor warms  
 tyre and floor together, and it absorbs differential drag too

# The Pack: L10.1, on Hardware You Can Hold

## EQUIVALENT-CIRCUIT MODEL AND STATE OF CHARGE

$$V_{\text{term}} = \text{OCV}(z) - iR_{\text{int}}, \quad \dot{z} = -\frac{i}{Q}$$

$z$  the state of charge,  $Q$  the capacity  
 $R_{\text{int}}$  the internal resistance

- **the simplest useful model** — an open-circuit voltage depending on state of charge
- in series with an internal resistance
- state of charge follows by coulomb counting
- an RC branch captures the relaxation transient, and is the first refinement worth adding
- this is L10.1's model, on hardware you can hold

### WHY SAG IS A CONTROL PROBLEM

Internal resistance connects L11.2 to the 5 W rule from L11.1. A hard acceleration pulls the terminal voltage down; if the compute rail is near its limit, the Nano browns out mid-run. The energy model and that reset are the same equation.

### THE PACK, SETTLED

**Four cells, 8.4 V, two in series and two in parallel.**

The series pair sets the voltage; the parallel pair doubles the capacity and roughly halves the internal resistance, so the pack sags less under hard acceleration.

Cells must be under 67 mm, so protected cells do not fit. A healthy 2S pack rests near 8.4 V; below that is a tired pack.



# The INA219: What It Actually Cost

## A SHUNT, AN AMPLIFIER, AND TWO VOLTAGES

$$i = \frac{V_{shunt}}{R_{shunt}}, \quad P = V_{bus} i$$

$V_{shunt}$  the voltage across the shunt  $R_{shunt}$   
 $V_{bus}$  the supply voltage

A small resistor sits in the supply line. The chip measures the voltage across it and the bus voltage, and current follows from Ohm's law.

Multiply and you have instantaneous power, over  $I^2C$ , from the car itself.

This is the component that makes the MiniProject an experiment. Without it you would compare a prediction against another prediction.

Configure it deliberately. The chip offers a choice of conversion time and on-chip averaging, and the defaults are chosen for slowly varying loads rather than for a PWM drive on a moving vehicle.

Read the whole car, not one branch. A shunt downstream of the compute rail measures traction only, and the hotel load decides the optimum.

*Record the sampling configuration in your log header.*

## THE ALIASING TRAP

The motor drive is switched at a few kilohertz. Sampling at 20 Hz with a short conversion window measures wherever in the PWM cycle the sample landed — noise dressed as data.

Set the conversion time and averaging so each reported sample integrates over many switching periods. Let the chip average, in hardware, before the value reaches you.

|                                     |                                |
|-------------------------------------|--------------------------------|
| <b>WHAT TO RECORD, EVERY SAMPLE</b> | monotonic clock, not wall time |
| <b><i>V, i, P</i></b>               | bus voltage, current, product  |
| <b><i>throttle,</i></b>             | the commands that were active  |
| <b><i>steering</i></b>              |                                |
| <b><i>v</i></b>                     | measured speed, if you have it |

# Turning Samples into an Energy Figure

## INTEGRATE POWER OVER THE RUN

$$E = \int_0^T Vi \, dt \approx \sum_k \frac{1}{2} (P_k + P_{k+1}) \Delta t_k$$

$P_k$  the power at sample  $k$   
 $\Delta t_k$  the time to the next

Trapezoidal integration over irregular timestamps, which is what you will have. The L11.1 loop jitters, and treating a 90 ms interval as 50 ms halves that step's energy.

Log from the control loop, not a separate thread. The current sample and the command that produced it must carry the same timestamp, or the identification has nothing to align.

Logging costs loop rate. Buffer in memory and write once at the end — an fsync inside the loop is the blocking call L11.1 warned about.

Report energy per lap and mean power separately. Quote only one and you cannot tell a fast efficient lap from a slow wasteful one.

## THE LOOP, EXTENDED FROM L11.1

- 1 **Capture** camera frame
- 2 **Infer** CNN forward pass
- 3 **Constrain** grip limit, hard layer
- 4 **Actuate** throttle and steering
- 5 **Sample** INA219: V, i, t
- 6 **Buffer** append, never write

*Steps 3, 5 and 6 are what L11.2 adds. Steps 5 and 6 must be cheap, or they will quietly undo the frame rate you fought for last week.*

# Fitting the Model to a Drive

## THIS IS AN INVERSE PROBLEM, AND YOU HAVE SOLVED ONE BEFORE

The structure of the model is known and four of its coefficients are not. That is exactly the inverse setting from L7: unknown parameters promoted to trainable variables, recovered by minimising a residual against measured data. The only difference is that the data came off a vehicle rather than a solver.

$\lambda = \{C_{rr}, Gk_t/r_w, R_a, R_{int}\}$  trainable

$C_{rr}$  rolling,  $Gk_t/r_w$  traction  
 $R_a$  armature,  $R_{int}$  pack resistance

Drive a rich excitation, not a demonstration lap. Steps, ramps, a coast-down, a steady cruise: a lap that only cruises leaves armature and rolling resistance indistinguishable.

The coast-down is the most informative manoeuvre you have. Cut the drive and log the deceleration: with no current the electrical terms vanish and the rolling coefficient falls out.

Scale the parameters first. Resistance is of order one ohm and capacity ten thousand coulombs; handed both at once, an optimiser moves on neither.

*Report the identified values with units and an honest uncertainty.*

## HOW TO KNOW WHETHER YOU BELIEVE THE FIT

### Hold out a run

Fit on one drive, predict another. A model that only reproduces its own training run has fitted the drive coefficients.

### Check the signs

A negative holding coefficient or a negative resistance is not a small numerical issue. It means the model is absorbing something you have not included.

### Compare magnitudes

Resistance of a few ohms, rolling coefficient of a few hundredths. Values orders away from these indicate a units error, usually in the gear ratio.

### Refit warm and cold

Winding resistance rises appreciably as the motors heat. If the two fits differ, that is physics, and it belongs in your report.

# The Loss, Assembled

## THREE TERMS, ONE OBJECTIVE

$$\mathcal{L}(\theta, \lambda) = \mathcal{L}_{\text{data}} + \lambda_{\text{dyn}} \mathcal{L}_{\text{dyn}} + \lambda_{\text{con}} \mathcal{L}_{\text{con}}$$

$\theta$  the weights,  $\lambda$  the parameters  
 $\lambda_{\text{dyn}}, \lambda_{\text{con}}$  the weights of the terms

$$\mathcal{L}_{\text{dyn}} = \frac{1}{N} \sum_{k=1}^N \|\dot{\mathbf{x}}(t_k) - \mathbf{f}(\mathbf{x}(t_k), \mathbf{u}(t_k); \lambda)\|^2$$

The data term anchors the trajectory to what the car did. The dynamics term penalises departure from the model. The constraint term handles what is not enforced by construction.

The residual's derivative is taken by autograd with respect to time — one input, not several. That is the whole difference from L7, and it makes the problem cheaper.

Weighting is the same difficulty as ever. Non-dimensionalise each residual before choosing  $\lambda$ , or the weights encode unit conversions.

*Everything above is L7 with one input instead of several.*

## WHAT CHANGED FROM L7

### Operator

#### $\partial/\partial t$ only

a system of ODEs, not a PDE

### Collocation

#### along a trajectory

no interior sampling of a domain

### Boundary data

#### an initial condition

one state, at  $t = 0$

### Unknowns

#### $\theta$ and $\lambda$ jointly

network weights and physical coefficients

### Ground truth

#### an instrument

a measured power trace, not a manufactured solution

# The Layer the Policy Cannot Override

The perception network from L11.1 proposes; it does not dispose. Between its output and the actuators sits a filter that knows the wheelbase, the friction limit and the speed.

This is not a safety afterthought bolted on. It is the hard-enforcement construction of slide 7, placed in the loop rather than the loss.

The filter must be cheap: a handful of scalar operations once per frame.

Give it authority over the throttle too. Steering saturation alone does not stop a car arriving at a corner too fast to take it.

And keep the L11.1 failsafe underneath. A late frame or a stale command still means stop.

*Write the filter before you write the policy.*

## COMMAND PATH, PROPOSAL TO ACTUATION

**CNN output** raw steering and throttle proposal

**Calibration map** command  $\rightarrow \delta$  in radians

**Curvature**  $\kappa = \tan \delta / L$ , hence  $R$

**Grip ceiling**  $v \leq v(\mu g R)$ , evaluated now

**Clamp** the binding limit wins

**Actuate** servo and motor commands

*Rows four and five are the whole of the safety argument, and together they are about six lines of code.*

# Asking for the Cheapest Lap

## ENERGY AND TIME IN ONE OBJECTIVE

$$J = \int_0^T [P(t) + w] dt = E + wT$$

$P$  the power,  $w$  the price of a second  
 $E$  the energy,  $T$  the lap time

Minimise energy alone and the answer is trivial: never move. Minimise time alone and you have the L11.1 objective with a flat battery. The weight  $w$  prices a second in joules.

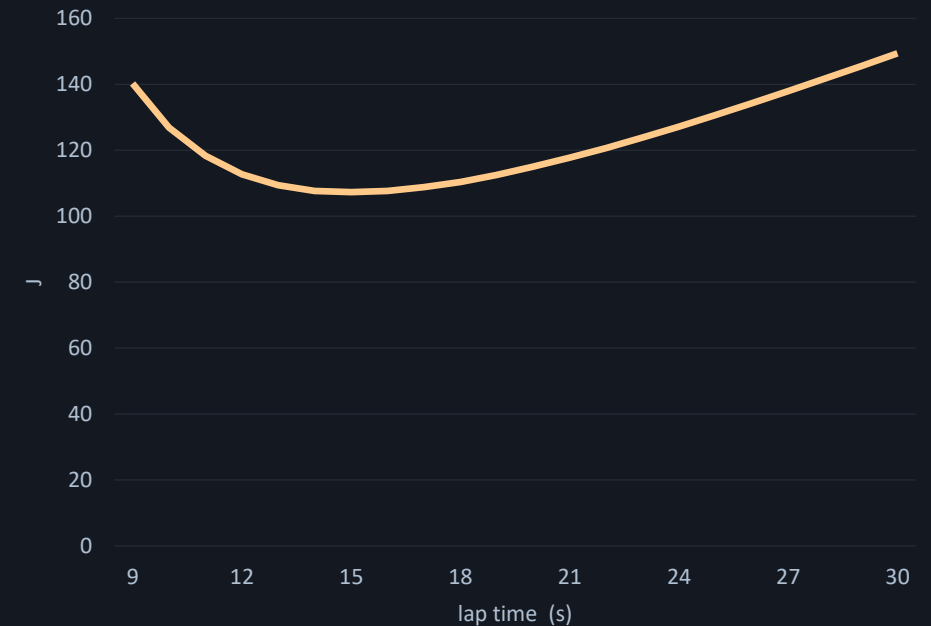
Subject to everything already established: the kinematics, the grip limit, the drivetrain and the pack.

Solve it directly. Discretise the lap, treat the speed in each segment as a variable, and hand it to a nonlinear programme.

### STATE $w$ EXPLICITLY IN YOUR REPORT

Two teams reporting different optimal speeds have usually chosen different  $w$  and not said so.

## ENERGY PER LAP AGAINST LAP TIME



Illustrative, with a 4.5 W hotel load over a 20 m lap. The curve has an interior minimum — and it is not at the slow end.



# Why the Efficient Lap Is the Smooth One

## THE CONVEXITY ARGUMENT

$$\overline{i^2} \geq \bar{i}^2 \quad \Rightarrow \quad \text{smooth effort costs less heat}$$

$i$  the motor current;  $\bar{\phantom{x}}$  the bar is the mean over the lap

Winding loss is quadratic in current, and current is proportional to tractive force. For a fixed mean force the mean square is smallest when the force is constant.

So a lap at steady effort dissipates less heat than one of equal average effort driven in bursts.

## AND THERE IS NO REGENERATION

$$E_{\text{lost}} = \frac{1}{2} m (v_{\text{in}}^2 - v_{\text{apex}}^2)$$

The kinetic energy carried into a corner and shed before the apex does not return to the pack. It becomes heat, and every joule was bought at the pack's efficiency.

Which gives the rule directly: arrive at the corner already at the speed the grip limit permits. Braking late is fast where energy is free. Here it is not.

## THE RESULT THE EXERCISE IS BUILT AROUND

The energy-optimal lap is the flattest speed profile that still respects the grip limit at every corner.

### Naïve fast

full throttle, brake at the corner —  
shortest time, highest energy

### Naïve slow

constant low speed throughout — lowest  
peak power, worst energy per lap

### Grip-limited smooth

speed set by upcoming curvature — near-  
minimum energy, close to fastest

# How These Models Fail

## **$\mu$ was measured yesterday**

Grip was identified on a clean floor and the demonstration runs after six other teams have shed dust and tyre rubber. Symptom: the car understeers wide at speeds that worked. Fix: re-measure at the start of every session.

## **The pack was not full**

Energy figures compared across runs at different states of charge. Internal resistance and terminal voltage both depend on it. Symptom: open-circuit voltage before each run and report it alongside the energy.

## **Parameters absorbed an error**

An uncalibrated steering map or a wrong gear ratio does not show up as a bad fit. The optimiser hides it in the rolling coefficient. Symptom: a good fit with a physically implausible coefficient. Fix: check magnitudes, not residuals.

## **Warm motors, cold model**

Winding resistance rises with temperature, so a model fitted on the first lap over-predicts efficiency on the tenth. Symptom: on a warm car, or let armature resistance depend on accumulated energy and identify that too.

## **The optimum was never tested**

A speed profile computed from an identified model, reported as the answer, and never driven on the floor. Fix: drive it and measure it. The whole point of the INA219 is that this claim is checkable.

*The first and third catch everyone. Neither is a modelling error — one is a measurement discipline failure and the other is an identifiability failure, and no amount of training fixes either.*

## Ex\_11.2 — Identify, Constrain, Optimise

### TASKS

- ① Measure  $L$  and calibrate the steering map over five commands in each direction.
- ② Instrument the loop: INA219 sampling, buffered logging, energy by trapezoidal integration.
- ③ Drive an excitation run — steps, ramps, a coast-down, a steady cruise — and log it.
- ④ Identify the rolling, traction, armature and internal-resistance parameters, then validate on a held-out run.
- ⑤ Add the grip-limit clamp as a hard layer beneath the L11.1 policy and confirm it binds.
- ⑥ Compute an energy-optimal speed profile, drive it, and compare predicted with measured energy.

*Runs on the same course as Ex\_11.1, with the same car and the same policy. What changes is that the car is now instrumented and the claims are now checkable.*

### QUESTIONS, AND WHERE ANSWERED

- |                                                                       |            |
|-----------------------------------------------------------------------|------------|
| Why does curvature, not yaw rate, follow from a steering command?     | sl. 5      |
| Why is a small excursion past the grip limit not an acceptable trade? | sl. 9      |
| Why can a slower lap consume more energy than a faster one?           | sl. 10, 18 |
| Why can the torque constant and gear ratio not be separated?          | sl. 11     |
| Why is the coast-down worth more than a demonstration lap?            | sl. 15     |
| Why is the smooth profile cheaper at equal average effort?            | sl. 19     |

# The Efficient Lap

## THE BRIEF

Complete a fixed course without collision, minimising energy per lap at a lap time you declare in advance. You choose  $w$ , you state it, and you defend it. Report predicted energy from your identified model alongside measured energy from the INA219, and account for the gap.

### Completes the course

No collision, no intervention, no stall. A run that does not finish does not score, whatever its power trace shows.

### Predicted vs measured

The headline number. Not the size of the error — the quality of the explanation for it.

### Constraint held

Evidence that the grip limit bound at least once and was never violated. A log excerpt, not an assertion.

### Pareto, not a point

At least three (time, energy) pairs from real runs, with  $w$  stated for each.

### Reproducible

Pack state, floor condition,  $\mu$ , and identified parameters recorded. Another team should be able to repeat it.

*A team that predicts 118 J, measures 141 J, and explains the difference as warm windings has done better physics than a team that predicts 118 J, measures 119 J, and cannot say why.*

Safety carries over unchanged from L11.1: one driver, one observer with a hand on the stop, throttle capped in software until the loop is measured, and the battery rules are not negotiable.

# Key Takeaways

- ① **Physics-informed does not mean PDE** Three ODEs — kinematics, grip, drivetrain — enter the loss exactly as a PDE residual did in L7. The operator changed; the machinery did not.
- ② **Enforce grip hard, not softly** A penalty is a preference the optimiser will trade away. A bounded layer scaled by  $v(\mu g R)$  cannot be traded, and on this vehicle the difference is a crash.
- ③ **Identify from excitation, not demonstration** Steps, ramps and above all a coast-down. A cruise leaves armature and rolling resistance indistinguishable, and the fit will look excellent anyway.
- ④ **The instrument is the point** The INA219 turns the MiniProject from a simulation into an experiment. For the first time in this course, the model can be wrong about something it cannot argue with.
- ⑤ **The cheapest lap is smooth, not slow** Quadratic winding loss plus a constant compute load put the energy optimum in the interior. Flatten the speed profile against the grip limit and stop braking energy you paid for.

*The car is still the benchmark. It now reports what it cost.*

# Where to Look

## Kong, Pfeiffer, Schildbach & Borrelli (2015)

Kinematic and dynamic vehicle models for autonomous driving control design. The standard reference for when the kinematic model is adequate and when it is not — read it for the validity argument, not the equations.

## Plett, Battery Management Systems, Vol. 1

Equivalent-circuit models, coulomb counting, and parameter identification. The source behind L10.1, and the place to go when your identified internal resistance misbehaves.

## Raissi, Perdikaris & Karniadakis (2019)

The PINN paper from L7. Section 4 is the inverse problem, which is exactly what slide 13 asks you to solve — with an ODE where they used a PDE.

*The vendor documentation remains the authority on the hardware; the academic references are the authority on the models. Where they disagree about this car, the hardware wins.*

## Rajamani, Vehicle Dynamics and Control

The textbook treatment: bicycle models, tyre forces, the friction circle. Chapters 2 and 13 cover everything used in this lecture, at more depth than the exercise needs.

## INA219 datasheet, Texas Instruments

Read the conversion-time and averaging registers specifically. The aliasing failure on slide 11 is entirely avoidable from this document.

## Waveshare JetRacer Pro AI Kit wiki

Still the authority on the hardware, and still the place to check the pack configuration before you trust any energy figure in this deck.

# Questions

## 1 – 2 · VEHICLE MODEL

### 1 What is the kinematic bicycle model, and what must be measured before it is used?

The car reduced to two wheels: speed and steering angle in, position and heading out, with the wheelbase  $L$  as its only parameter. The map from steering command to angle is not linear and is calibrated by measurement, or every later result is in doubt. *Ex\_11.2 notebook 10*

### 2 What is the grip limit, and why is it enforced hard?

The tyres give one budget of force, which braking and cornering share, so the speed in a bend of radius  $R$  cannot pass  $v(\mu g R)$ . A penalty is a preference the optimiser may trade away; a bounded layer cannot be traded, and on a car the difference is a crash. *Ex\_11.2 notebooks 10, 20; L7.1*

## 3 – 4 · ENERGY

### 3 Where does the energy go, and why is the cheapest lap not the slowest?

Into winding loss, which grows with the square of the current, into rolling resistance, and into the computer, which draws a constant power for as long as the lap takes. A slow lap pays the constant load longer, so the optimum lies in between, at a smooth speed profile. *Ex\_11.2 notebooks 10, 20*

### 4 How are the model's parameters identified, and what makes the result checkable?

From an excitation run with steps, ramps and a coast-down; a steady cruise cannot separate the armature resistance from the rolling resistance, though the fit looks good. The current sensor measures the energy of a lap, so the prediction is compared with a measurement. *Ex\_11.2 notebook 10; L10.1*



# Energy Optimisation of Navigation with the JetRacer

## THE CAR AS DELIVERED



### 4WD

all four wheels driven  
front and rear, shaft-

### differentials

linked

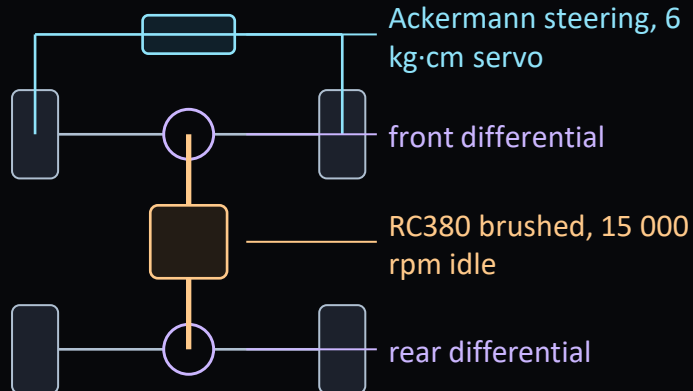
### 8.4 V

four 18650s, 2S2P

### no encoder

nothing measures speed

## DRIVELINE



*no encoder on any axle — speed must be timed or inferred*

## WHAT MATTERS FOR ENERGY

### Compute

Jetson Nano 4 GB, 5 W mode

### Pack

4 × 18650, 8.4 V, 2S2P

### Motors

RC380 brushed, 15 000 rpm

### Transmission

front and rear differentials

### Power monitor

INA219 over I<sup>2</sup>C, bus + shunt

### Hotel load

a few watts, always on

### Speed sensing

none fitted — no encoder

### Wheelbase

measure it yourself

*Waveshare, Pro column.*

## WHAT YOU BUILD ON TOP OF IT

- ① **Instrument**  
INA219 in the loop, buffered log, trapezoidal energy
- ② **Identify**  
coast-down and excitation — no encoder, so time it
- ③ **Constrain**  
grip-limit clamp as a hard layer, not a penalty
- ④ **Optimise**  
speed profile minimising  $E + wT$ , then drive it

## WHERE YOU START FROM

Your own Ex\_11.1 policy, driving the course at one fixed throttle.

It has no idea what it costs. It cannot slow for a corner, it cannot spend less on a straight, and it treats the pack as infinite.

Nothing in it is wrong. It is simply optimising one axis of a two-axis problem.

## THE NUMBER THAT DECIDES IT

Predicted energy against measured energy, on a lap you computed before you drove it.

# The Efficiency Trophy — Six Cars, Two Leaderboards

## THE WEIGHT IS FIXED, AND ANNOUNCED BEFORE YOU START

Every team optimises the same cost functional with  $w = 5 \text{ J/s}$  — one second of lap time priced at five joules. You may not choose your own  $w$  for the scored run.

### BOARD A — EFFICIENCY

$$\eta = \frac{J_0}{J}, \quad J = E_{\text{lap}} + w T_{\text{lap}}$$

$J$  is the scored lap's cost;  $J_0$  is the same car's baseline from the stock policy.  $\eta > 1$  means you beat the car you were given.

Normalising per car ranks the teams, not the hardware.

### BOARD B — MODEL FIDELITY

$$\varphi = \frac{E_{\text{meas}}}{E_{\text{meas}} + |E_{\text{pred}} - E_{\text{meas}}|}$$

Predicted energy comes from your model and your computed speed profile, committed in writing before the run. Measured energy comes from the INA219.

A team can win Board A by driving well and still finish last here.

## GATES — FAIL ANY ONE AND THE RUN DOES NOT SCORE

- **Completes** three consecutive laps, no contact, no intervention
- **Clamp binds** log shows the grip limit binding at least once
- **Clamp holds** log shows lateral acceleration never exceeding  $\mu g$
- **Committed** predicted energy timestamped before the run

### THE TROPHY

Champion is the lowest rank sum across both boards. A tie goes to Board B.